BS-M301 — Probability and Statistics

Semester III • 3L:0T:3P • 3 Credits

Module 5 Exam Notes: Applied Statistics, Curve Fitting and Tests of Significance

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

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1. Curve Fitting by the Method of Least Squares

In experimental electronics and engineering measurement, observational data points $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ rarely lie on an exact mathematical curve due to sensor noise and environmental fluctuations. The **Method of Least Squares** provides an objective criterion to establish the best-fitting functional relationship $y = f(x; a, b, c, \dots)$ by minimizing the sum of the squared vertical discrepancies.

Definition: The Principle of Least Squares

Let (x_i, y_i) be n paired experimental measurements. For any chosen curve $y = f(x)$, the residual (vertical error) for the i-th observation is:

$$e_i = y_i - f(x_i) \tag{1}$$

The Principle of Least Squares states that the optimal curve parameters are those that minimize the total sum of squared residuals:

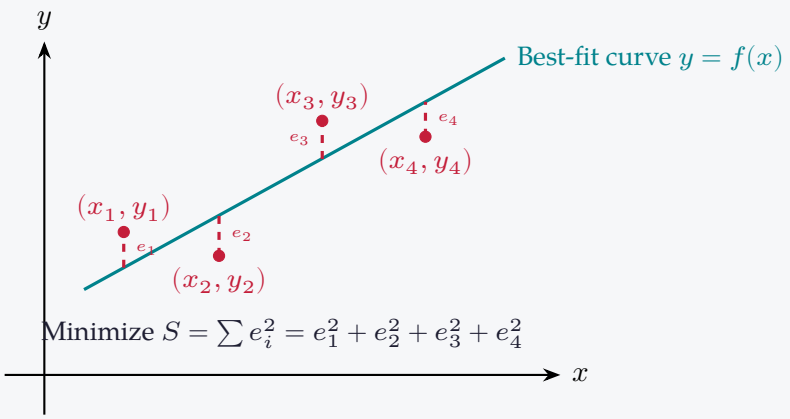
$$S = \sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} [y_i - f(x_i; a, b, c, \dots)]^2 = \text{Minimum} \tag{2}$$

To achieve this minimum, the partial derivatives of S with respect to each undetermined parameter must vanish simultaneously:

$$\frac{\partial S}{\partial a} = 0, \qquad \frac{\partial S}{\partial b} = 0, \qquad \frac{\partial S}{\partial c} = 0, \quad \dots \tag{3}$$

The resulting system of simultaneous equations is called the **Normal Equations**.

TikZ Schematic: Residual Errors and Best-Fit Curve in Least Squares



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1.1 1. Fitting of a Straight Line: $y = a + bx$

Derivation of Normal Equations for a Straight Line

Let the straight line equation be $y = a + bx$. The error sum of squares is:

$$S = \sum_{i=1}^{n} (y_i - a - bx_i)^2 \tag{4}$$

Differentiating partially with respect to a and equating to zero:

$$\frac{\partial S}{\partial a} = \sum_{i=1}^{n} 2(y_i - a - bx_i)(-1) = 0 \implies \sum_{i=1}^{n} (y_i - a - bx_i) = 0$$
$$\implies \sum y = na + b \sum x \tag{5}$$

Differentiating partially with respect to b and equating to zero:

$$\frac{\partial S}{\partial b} = \sum_{i=1}^{n} 2(y_i - a - bx_i)(-x_i) = 0 \implies \sum_{i=1}^{n} x_i(y_i - a - bx_i) = 0$$
$$\implies \sum xy = a \sum x + b \sum x^2 \tag{6}$$

The Normal Equations for a Straight Line:

$$\sum y = na + b \sum x \tag{7}$$
$$\sum xy = a \sum x + b \sum x^2 \tag{8}$$

Computation Shortcut (Change of Origin): If n is odd and values of x are equally spaced, choosing the origin at the midpoint such that $\sum u = 0$ simplifies the solution to:

$$a = \frac{\sum y}{n}, \qquad b = \frac{\sum uy}{\sum u^2} \tag{9}$$

1.2 2. Fitting of a Second-Degree Parabola: $y = a + bx + cx^2$

Normal Equations for a Second-Degree Parabola

For the quadratic curve $y = a + bx + cx^2$, the sum of squared errors is $S = \sum(y_i - a - bx_i - cx_i^2)^2$. Setting $\frac{\partial S}{\partial a} = 0$, $\frac{\partial S}{\partial b} = 0$, $\frac{\partial S}{\partial c} = 0$ yields the 3×3 system of **Normal Equations**:

$$\sum y = na + b \sum x + c \sum x^2 \tag{10}$$
$$\sum xy = a \sum x + b \sum x^2 + c \sum x^3 \tag{11}$$
$$\sum x^2 y = a \sum x^2 + b \sum x^3 + c \sum x^4 \tag{12}$$

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When origin is shifted to make $\sum u = 0$ (which also implies $\sum u^3 = 0$ for symmetric spacing):

$$\sum y = na + c\sum u^2 \tag{13}$$

$$\sum uy = b\sum u^2 \implies b = \frac{\sum uy}{\sum u^2} \tag{14}$$

$$\sum u^2 y = a\sum u^2 + c\sum u^4 \tag{15}$$

1.3 3. Fitting of Non-Linear / General Curves via Linearization

Many non-linear physical relationships can be linearized by taking logarithms:

Curve Model	Log Transformation	Transformed Form	Normal Equations
Exponential Curve $y = ae^{bx}$	$\ln y = \ln a + bx$	$Y = A + bx$ ($Y = \ln y, A = \ln a$)	$\sum Y = nA + b\sum x$ $\sum xY = A\sum x + b\sum x^2$
Power / Geometric $y = ax^b$	$\log_{10} y = \log_{10} a + b\log_{10} x$	$Y = A + bX$ ($Y = \log y, X = \log x$)	$\sum Y = nA + b\sum X$ $\sum XY = A\sum X + b\sum X^2$
Modified Exponential $y = ab^x$	$\log y = \log a + x\log b$	$Y = A + Bx$ ($A = \log a, B = \log b$)	$\sum Y = nA + B\sum x$ $\sum xY = A\sum x + B\sum x^2$
Hyperbolic Curve $y = \frac{a}{x} + b$	**Substitute** $u = \frac{1}{x}$	$y = au + b$	$\sum y = a\sum u + nb$ $\sum uy = a\sum u^2 + b\sum u$

2. Foundations of Statistical Hypothesis Testing

Definition: Terminology of Statistical Testing

1. **Statistical Hypothesis:** An assertion or conjecture concerning the unknown parameter(s) of a population distribution.
2. **Null Hypothesis (H_0):** The baseline hypothesis of *no effect*, *no difference*, or *status quo*. It is maintained until evidence indicates otherwise.
3. **Alternative Hypothesis (H_1):** The mutually exclusive claim that contradicts H_0. It can be **two-tailed** ($H_1 : \theta \neq \theta_0$) or **one-tailed** ($H_1 : \theta > \theta_0$ or $H_1 : \theta < \theta_0$).
4. **Errors in Hypothesis Testing:**
   - **Type I Error (α):** Rejecting H_0 when H_0 is actually true (Producer's risk / False

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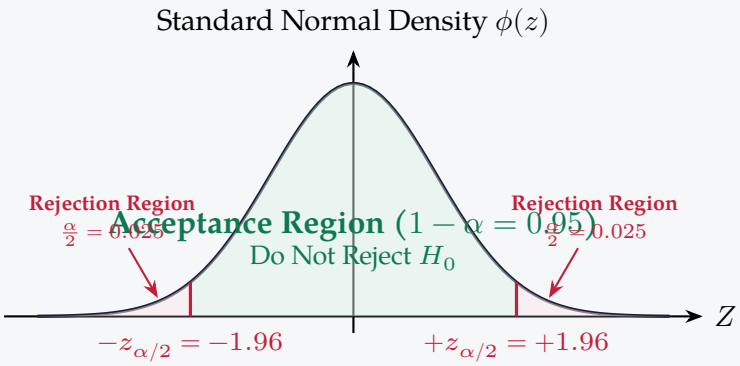
positive).

$$P(\text{Type I Error}) = \alpha = \text{Level of Significance} \tag{16}$$

- **Type II Error (β):** Failing to reject H_0 when H_0 is actually false (Consumer's risk / False negative).
- **Power of the Test ($1-\beta$):** Probability of correctly rejecting a false null hypothesis.

5. **Critical Region (Rejection Region):** The set of values of the test statistic for which H_0 is rejected.

TikZ Schematic: Standard Normal Test Statistic with Two-Tailed Critical Regions ($\alpha = 0.05$)



2.1 Critical Values for Large Sample Tests (Z-Distribution)

Level of Significance (α)	**Two-Tailed Critical Value ($\lvert Z\rvert > z_{\alpha/2}$)**	**Right-Tailed Critical Value ($Z > z_{\alpha}$)**	**Left-Tailed Critical Value ($Z < -z_{\alpha}$)**
$\alpha = 5\%$ (0.05)	$z_{0.025} = 1.96$	$z_{0.05} = 1.645$	$-z_{0.05} = -1.645$
$\alpha = 1\%$ (0.01)	$z_{0.005} = 2.576$	$z_{0.01} = 2.326$	$-z_{0.01} = -2.326$
$\alpha = 10\%$ (0.10)	$z_{0.05} = 1.645$	$z_{0.10} = 1.282$	$-z_{0.10} = -1.282$

3. Large Sample Tests of Significance (Z-Tests, $n \geq 30$)

By the Central Limit Theorem, for sample sizes $n \geq 30$, sample statistics (proportions, means, variances) are approximately normally distributed regardless of the underlying parent population distribution. The general test statistic is:

$$Z = \frac{\text{Sample Statistic} - \text{Hypothesized Parameter Value}}{\text{Standard Error of the Statistic (SE)}} \sim \mathcal{N}(0,1) \tag{17}$$

3.1 1. Large Sample Test for a Single Proportion

Testing whether a population proportion P equals a specified value P_0.

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Test for Single Proportion

- **Hypotheses:** $H_0 : P = P_0$ vs $H_1 : P \neq P_0$ (two-tailed) or $P > P_0$ (right-tailed).
- **Sample Proportion:** $p = \frac{x}{n}$, where x is the number of successes in n trials.
- **Standard Error under H_0:** $\text{SE}(p) = \sqrt{\frac{P_0 Q_0}{n}}$, where $Q_0 = 1 - P_0$.
- **Test Statistic:**

$$Z = \frac{p - P_0}{\sqrt{\frac{P_0 Q_0}{n}}} \tag{18}$$

- **Decision Rule:** If $|Z| > 1.96$ (at 5% level), reject H_0.

3.2 2. Large Sample Test for Difference of Two Proportions

Testing whether two independent populations have identical success proportions ($P_1 = P_2$).

Test for Difference of Two Proportions

- **Hypotheses:** $H_0 : P_1 = P_2$ vs $H_1 : P_1 \neq P_2$.
- **Sample Proportions:** $p_1 = \frac{x_1}{n_1}$ and $p_2 = \frac{x_2}{n_2}$.
- **Pooled Proportion Estimate (under H_0):**

$$\hat{p} = \frac{x_1 + x_2}{n_1 + n_2} = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2}, \qquad \hat{q} = 1 - \hat{p} \tag{19}$$

- **Test Statistic:**

$$Z = \frac{p_1 - p_2}{\sqrt{\hat{p}\hat{q}\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} \tag{20}$$

3.3 3. Large Sample Test for a Single Mean

Testing whether a population mean μ equals a specified value μ_0.

Test for Single Mean

- **Hypotheses:** $H_0 : \mu = \mu_0$ vs $H_1 : \mu \neq \mu_0$.
- **Sample Mean:** $\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$.
- **Test Statistic:**

$$Z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} \tag{21}$$

If the population standard deviation σ is unknown, it is replaced by sample SD $s = \sqrt{\frac{1}{n}\sum(x_i - \bar{x})^2}$ (valid since $n \geq 30$):

$$Z = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} \tag{22}$$

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3.4 4. Large Sample Test for Difference of Two Means

Testing whether two independent populations have equal means ($\mu_1 = \mu_2$).

Test for Difference of Two Means

- **Hypotheses:** $H_0 : \mu_1 = \mu_2 \quad \text{vs} \quad H_1 : \mu_1 \neq \mu_2$.
- **Test Statistic:**

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \approx \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} \tag{23}$$

3.5 5. Large Sample Test for Difference of Standard Deviations

Testing whether two populations have identical variability ($\sigma_1 = \sigma_2$).

Test for Difference of Two Standard Deviations

For a sample of size n drawn from a normal population, the sampling distribution of the sample standard deviation s is asymptotically normal with mean σ and standard error:

$$\text{SE}(s) = \frac{\sigma}{\sqrt{2n}} \tag{24}$$

Test Statistic for $H_0 : \sigma_1 = \sigma_2$:

$$Z = \frac{s_1 - s_2}{\sqrt{\frac{\sigma_1^2}{2n_1} + \frac{\sigma_2^2}{2n_2}}} \approx \frac{s_1 - s_2}{\sqrt{\frac{s_1^2}{2n_1} + \frac{s_2^2}{2n_2}}} \tag{25}$$

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Quick Revision — Module 5

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Key Normal Equations and Large Sample Z-Test Formulas at a Glance

Line Normal Equations	$\sum y = na + b\sum x, \quad \sum xy = a\sum x + b\sum x^2$
Parabola Normal Eqs.	$\sum y = na + b\sum x + c\sum x^2, \quad \sum xy = a\sum x + b\sum x^2 + c\sum x^3, \quad \sum x^2 y = a\sum x^2 + b\sum x^3 + c\sum x^4$
Exponential Curve	$y = ae^{bx} \implies \ln y = \ln a + bx \implies Y = A + bx$
Power Curve	$y = ax^b \implies \log y = \log a + b\log x \implies Y = A + bX$
Single Proportion Test	$Z = \frac{p - P_0}{\sqrt{P_0 Q_0/n}}, \quad Q_0 = 1 - P_0$
Two Proportions Test	$Z = \frac{p_1 - p_2}{\sqrt{\hat{p}\hat{q}(1/n_1 + 1/n_2)}}, \quad \hat{p} = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2}, \ \hat{q} = 1 - \hat{p}$
Single Mean Test	$Z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} \approx \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$
Two Means Test	$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}} \approx \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{s_1^2/n_1 + s_2^2/n_2}}$
Standard Deviation SE	$\mathrm{SE}(s) = \frac{\sigma}{\sqrt{2n}}, \quad Z = \frac{s_1 - s_2}{\sqrt{s_1^2/(2n_1) + s_2^2/(2n_2)}}$
Critical Z-Values	$\alpha = 5\% : \lvert Z\rvert = 1.96$ (two-tailed), 1.645 (one-tailed); $\alpha = 1\% : \lvert Z\rvert = 2.576$

One-Line Definitions

- **Least Squares Principle:** Criterion minimizing the sum of squared vertical residuals $\sum(y_i - f(x_i))^2$.
- **Normal Equations:** Set of simultaneous linear equations obtained by setting partial derivatives of error to zero.
- **Null Hypothesis (H_0):** Initial baseline claim stating no difference, no effect, or parameter equality.
- **Alternative Hypothesis (H_1):** The competing research claim adopted if sample evidence contradicts H_0.
- **Type I Error (α):** Mistakenly rejecting a true null hypothesis (false positive; significance level).
- **Type II Error (β):** Incorrectly failing to reject a false null hypothesis (false negative).
- **Power of Test ($1 - \beta$):** Probability of correctly rejecting H_0 when the alternative H_1 is true.
- **Critical Region:** Subspace of test statistics leading to rejection of H_0 at significance level α.
- **Standard Error (SE):** Standard deviation of the sampling distribution of a given sample statistic.
- **Pooled Proportion ($\hat{p}$):** Combined success rate $(x_1 + x_2)/(n_1 + n_2)$ under the null hypothesis of equality.

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Frequently Asked / Viva Questions (Part 1)

Q1. State the Principle of Least Squares and explain why vertical errors are squared.

The principle minimizes $S = \sum e_i^2 = \sum (y_i - \hat{y}_i)^2$. Errors are squared because (1) positive and negative errors would otherwise cancel out ($\sum e_i = 0$), (2) squaring penalizes large outliers heavily, and (3) S is continuously differentiable, yielding linear normal equations.

Q2. How can the exponential curve $y = ab^x$ be fitted using the method of least squares?

Taking base-10 logarithms: $\log y = \log a + x \log b$. Defining $Y = \log y$, $A = \log a$, and $B = \log b$, the model becomes the linear form $Y = A + Bx$. Solving standard linear normal equations for A and B, the parameters are recovered via $a = 10^A$ and $b = 10^B$.

Q3. Distinguish between Type I Error and Type II Error in hypothesis testing.

Type I error (α) occurs when a true null hypothesis H_0 is mistakenly rejected (false alarm). Type II error (β) occurs when a false null hypothesis is accepted (missed detection). Decreasing α for a fixed sample size increases β.

Q4. Why can the normal distribution be used for large sample tests even if the population is non-normal?

By the Central Limit Theorem (CLT), the sampling distribution of the sample mean $\bar{x}$ (and sample proportions) approaches a normal distribution as sample size $n \geq 30$, irrespective of the shape of the parent population.

Q5. What is the difference between a one-tailed test and a two-tailed test?

A two-tailed test investigates departures in either direction ($H_1 : \mu \neq \mu_0$), placing critical area $\alpha/2$ in both tails. A one-tailed test investigates directional superiority or inferiority ($H_1 : \mu > \mu_0$ or $\mu < \mu_0$), concentrating the entire critical area α in one tail.

Q6. State the standard error of the sample standard deviation for a normal population.

For large samples drawn from a normal population, $\text{SE}(s) = \frac{\sigma}{\sqrt{2n}}$. It is used to test significance of variance differences across industrial production lines.

Q7. Why do we pool the proportions $\hat{p} = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2}$ in the two-proportion test?

Under the null hypothesis $H_0 : P_1 = P_2 = P$, both samples originate from a single population with common proportion P. Pooling gives the minimum-variance unbiased estimator of this common parameter.

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Frequently Asked / Viva Questions (Part 2)

Q8. What decision is made if the calculated test statistic is $Z_{\text{cal}} = 2.15$ at 5% significance level?

For a two-tailed test at $\alpha = 5\%$, the critical boundary is $Z_{\text{crit}} = 1.96$. Since $|Z_{\text{cal}}| = 2.15 > 1.96$, the sample statistic falls in the critical rejection region. We reject the null hypothesis H_0 and conclude the observed difference is statistically significant.

Q9. How does shifting the origin simplify fitting a parabola $y = a + bx + cx^2$?

If x values are symmetrically spaced, shifting the origin to the center makes $\sum u = 0$ and $\sum u^3 = 0$. The normal equations decouple into $b = \sum uy / \sum u^2$, while a and c decouple from b into an elementary 2×2 system, dramatically reducing computational effort.

Q10. What is the Level of Significance (α) and how is it related to the confidence level?

The level of significance α is the maximum tolerable probability of committing a Type I error. It is related to the confidence level C by $C = (1 - \alpha) \times 100\%$. For example, an $\alpha = 0.05$ level of significance corresponds to a 95% confidence level.

Summary Matrix: Comprehensive Decision Table of Large Sample Significance Tests

Test Type	Null Hypothesis (H_0)	Standard Error (SE)	Test Statistic (Z_{cal})
Single Proportion	$H_0 : P = P_0$	$\sqrt{\dfrac{P_0 Q_0}{n}}$	$Z = \dfrac{p - P_0}{\sqrt{P_0 Q_0 / n}}$
Two Proportions	$H_0 : P_1 = P_2$	$\sqrt{\hat{p}\hat{q}\left(\dfrac{1}{n_1} + \dfrac{1}{n_2}\right)}$	$Z = \dfrac{p_1 - p_2}{\sqrt{\hat{p}\hat{q}(1/n_1 + 1/n_2)}}$ where $\hat{p} = \dfrac{n_1 p_1 + n_2 p_2}{n_1 + n_2}$, $\hat{q} = 1 - \hat{p}$
Single Mean	$H_0 : \mu = \mu_0$	$\dfrac{\sigma}{\sqrt{n}} \approx \dfrac{s}{\sqrt{n}}$	$Z = \dfrac{\bar{x} - \mu_0}{s/\sqrt{n}}$
Two Means	$H_0 : \mu_1 = \mu_2$	$\sqrt{\dfrac{s_1^2}{n_1} + \dfrac{s_2^2}{n_2}}$	$Z = \dfrac{\bar{x}_1 - \bar{x}_2}{\sqrt{s_1^2/n_1 + s_2^2/n_2}}$
Two Std. Deviations	$H_0 : \sigma_1 = \sigma_2$	$\sqrt{\dfrac{s_1^2}{2n_1} + \dfrac{s_2^2}{2n_2}}$	$Z = \dfrac{s_1 - s_2}{\sqrt{s_1^2/(2n_1) + s_2^2/(2n_2)}}$

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